

ProbOS — Month 3, Week 14

Injection Molding Process Qualification (Research Gate First)

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Week 14 goal

Add injection molding process qualification as a new domain model, reusing the existing, validated engine (`Distribution/ Model ABCs`, `MonteCarloEngine`, `SobolSensitivity`, the now-fixed reproducible seeding, C++ `LogNormal/ Uniform` classes, PDSL v0.2) — explicitly NOT a new architecture, a new model class on infrastructure already at 408/408 green tests.

Decision, stated up front: build the real model first, with a genuine research gate (Day 1, before any model code), THEN add it as a fourth tab to the live demo — not assuming the originating pitch document’s claim that “public literature on PP/PC process windows” exists without verifying it directly.

Day 1 – Research gate (before any model code)

- Search for a real, citable, published injection molding process-window dataset (PP or PC material) with real reported process parameters and real reported out-of-spec/defect outcomes — the exact discipline already used for `PharmaStabilityModel` (Gonzalez-Gonzalez et al. 2023)
- **Explicit decision gate:** if real, usable data is found, proceed to Day 2 as planned below. If not found, document the gap honestly and either narrow the model’s scope to what real data DOES support, or defer the week’s remaining time to another already-planned item

Day 2 – InjectionMoldingProcessModel (contingent on Day 1)

- Same architectural pattern as `BatteryModel2Cell`: state variables (cavity pressure, melt temperature, fill time, cooling time, part dimensional outcome), parameters with real uncertainty informed by Day 1’s findings (resin lot viscosity — reuses the existing C++ `LogNormal` class directly; ambient humidity and machine repeatability as `Normal`; wall- thickness tolerance from mold wear)
- Output: probability of an out-of-spec part as a function of the process window — directly analogous to the battery model’s thermal-runaway probability output

Day 3 – Monte Carlo + Sobol sensitivity

- Wire the real `MonteCarloEngine` and `SobolSensitivity` (now genuinely reproducible, per this session’s own seeding-bug fix and its permanent regression test) to the new model

- The actual sellable insight: “of N process parameters, which 2–3 are actually driving scrap rate”

Day 4 – Validation against the real Day 1 dataset

- Validate against whatever real dataset Day 1 found, with an honestly stated tolerance and methodology strength

Day 5 – PDSL spec for process windows

- A real PDSL v0.2 example letting a process engineer define a process window declaratively

Day 6 – Fourth demo tab

- Add Injection Molding as a fourth tab to the live demo, following the EXACT same discipline already established: real pre-computed kernel data, no JS toy-formula reimplementations, real interpretation bands, real Sobol breakdown, hover tooltips

Day 7 – Audit, documentation, retrospective

- `pre_week_audit.sh`, full test suite, CI, retrospective

Business framing (for reference, not this week’s engineering scope)

Open-core (Apache 2.0) for the model and Sobol analysis; a paid audit-trail tier reframes the earlier MoldMind + ValidTrail ideas as one feature on existing infrastructure. Outreach is explicitly OUT of this week’s scope.

Exit criteria

A genuinely validated (or honestly gap-documented) injection molding domain model, wired to the real Sobol/Monte Carlo engine, with a working PDSL example and a fourth live demo tab built on real data.